Mohamed Bahgat *Teaching Assistant / AI Engineer*

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Summary

Teaching Assistant in the Artificial Intelligence Engineering Department at El Alamein International University, with a focus on deep learning, machine learning, reinforcement learning, and NLP. Currently leading AI Research and Development at VRtualize Srl, contributing to impactful projects in healthcare and finance. Passionate about advancing knowledge in AI through both academic and practical applications, and eager to deepen expertise in the field through the Master's program in Artificial Intelligence at Nile University.

Professional Experience

2024/08–present
Milan, Italy
Remote

AI Research and Development Team Lead
VRtualize Srl (Italy)

- Lead a multidisciplinary team of 12 AI and software researchers to develop cutting-edge AI solutions.
- Led AI R&D initiatives, driving innovation in healthcare and finance through cutting-edge machine learning solutions.
- Coordinated cross-functional teams to develop AI-driven applications, improving operational efficiency and decision-making.
- Published research findings and contributed to the adoption of AI solutions in diverse fields.
- Collaborate with European corporations and Italian research laboratories to design innovative solutions for complex challenges in healthcare and beyond.

2024/09–present
Alamein, Matrouh

Teaching Assistant – Artificial Intelligence Engineering Dept.
El Alamein International University

- Taught advanced AI topics, including deep learning, machine learning, reinforcement learning, and NLP, to undergraduate students.
- Designed and delivered interactive course materials to enhance student engagement and understanding of complex AI concepts.

2024/07 – 2024/10

AI Engineer at Horeca smart company

- Working on the analysis and forecasting for the sales team using Microsoft Power BI and Amazon Sage .
- Working on developing chatbots using long-chain,botpress

Internships

2024/05–2024/11
Cairo

Digital Egypt Pioneers Initiative - DEPI (Generative AI Track)
This training is under the guidance Ministry of Communications and Information Technology (MCIT), The main courses: ML , DL, Mlops Tools, Hugging Face, Variant Types of GANS, NLP With Attention, Gen AI

Education

Bachelor of Computer Science and Artificial Intelligence
Faculty of Computer Science and Artificial Intelligence, Banha University
Graduation Year: [2024] CGPA: 3.30 / 4.00 (Very Good with Honor)
Graduation Project: Automatic Grading for Essay Questions Using Deep Learning Models

- Developed an AI-driven system to automate essay grading, reducing manual correction time, minimizing human error, and optimizing evaluation costs.
- Utilized advanced language models (LLAMA, MISTRAL, and FLAN T5) to understand semantic meaning, achieving 97% accuracy on an English dataset and 92% accuracy on an Arabic dataset.

Technical Skills

- Data Analysis (Pandas, Matplotlib, SciPy, Plotly, PowerPI, Signal and Text analysis).
- Software Development (C++, Python, OOP, SQL, Flask, Data Structures, Algorithm, Git, GitHub, Databases, Linux).
- Applied Mathematics (Calculus, Linear Algebra, Probability, Statistics).
- Machine and deep learning (Sikit-Learn, Pytorch, TensorFlow, Spacy, Cuda, Data Modeling, Model Optimization, and Evaluation, Generative Modelling, Transformers, Sequence Models, Reinforcement Learning).

Soft Skills

- Problem-solving & Analytical Thinking: Adept at translating complex data into actionable insights, with experience in model optimization and evaluation.
- Project Management: Demonstrated ability to balance multiple roles, managing AI research projects, academic responsibilities, and client relationships.
- Team Leadership & Collaboration: Led a cross-functional team of 12 researchers, driving innovative AI solutions while fostering a collaborative environment.

Project

Arabic Meeting Summarization | June 2024 – Oct 2024

- Developed a summarization system for Arabic meetings by translating datasets from Dialogsum.
- Leveraged **Bloomz 3b** and **LoRA** technologies to enhance summarization accuracy. Successfully processed Arabic datasets through efficient translation methods

Age Prediction from Facial Images | June 2024 – Aug 2024

- Developed a Convolutional Neural Network (CNN) model for age prediction using facial images, achieving an R^2 score of 0.70 through optimized model architecture and preprocessing techniques.

Machine Translation: English to Arabic | Feb 2024 – May 2024

- Designed a Seq2Seq MT model based on the Transformer for translating English to Arabic.
- Achieved a BLEU score of 88, demonstrating high translation accuracy. Conducted human evaluation to validate results, achieving precision compared to benchmark translations.

Speech Emotion Detector | Feb 2023 – May 2023

- Developed a system using Convolutional Neural Networks (CNNs) to analyze pre-recorded human audio and identify the speaker's emotions based on the sound.

Chatbot Development Using Transformer | Feb 2024 – May 2024

- Designed and implemented a chatbot from scratch using the Transformer architecture to answer users.
- Trained the model on the **IMDB dataset**, leveraging movie reviews to build a conversational system capable of responding accurately based on user input.

Languages

Arabic —Native **English** — Very Good